Reduced Flood Model Variable Tree

The 12 inputs and 2 controls of the narrowed impact model, one per physical mechanism. The same set feeds both the GLM hurdle and the LightGBM hurdle. The 27 dropped inputs and the unused panel columns are listed after the tree.

Hazard (W) · Long-run rain climate (C) · Exposure and susceptibility (X) · Adaptation (A) · Controls

Y: county-month property loss

`has_damage` (part 1) × `log_PropertyDmgADJ` given damage (part 2) · SHELDUS, 2024 $

Hazard (W)

6 inputs · nClimGrid-Daily, ERA5-Land

SAME MONTH

- **`p_month`** — Monthly rain total (log)
- **`rx5_rl_ratio`** — Wettest 5 days ÷ county 10-yr level
- **`snowmelt_mm`** — Snowmelt in the month (log)
- **`frost_days`** — Days with Tmin < 0 °C (frozen ground)

KNOWN IN ADVANCE (DAY 1)

- **`sm_root_pct`** — Root-zone soil wetness percentile
- **`swe_mm`** — Snowpack water equivalent (log)

Rain climate (C)

1 input · nClimGrid-Daily, 1951–1999

- **`c_rx5_rl10_mm`** — 10-year 5-day rain level (confounder)

Exposure & susceptibility (X)

4 inputs · NHGIS, NLCD, MERIT Hydro, TIGER

- **`log_housing_units`** — Homes exposed
- **`impervious_mean`** — Paved share (runoff), year − 1
- **`hand_lt5_share`** — Land < 5 m above drainage
- **`log_land_km2`** — County land area (scale)

Adaptation (A)

1 input · Annual NLCD

- **`wetland_share`** — Natural water storage, year − 1

Controls

2 fixed effects

- **`state`** — Reporting differences
- **`calendar month`** — Seasonality

Dropped from the models

Struck-through columns stay in `model_panel.csv` but are no longer model inputs. Each repeats a kept variable (in parentheses).

W: rain duplicates

~~`rx1`~~, ~~`rx3`~~, ~~`rx5`~~ (rx5_rl_ratio)

~~`n_gt_p95`~~, ~~`n_gt_p99`~~ (p_month)

~~`rx5_gt_rl10`~~ (rx5_rl_ratio)

~~`rain_x_wetsoil`~~ (interaction)

W: soil and snow duplicates

~~`sm_surface`~~, ~~`sm_root`~~, ~~`sm_root_anom`~~ (sm_root_pct)

~~`spi1_prev`~~, ~~`spi2_prev`~~ (sm_root_pct)

~~`snowmelt_prev_mm`~~ (swe_mm)

~~`frost_days_prev`~~ (frost_days)

C: other thresholds

~~`c_wetday_p95_mm`~~, ~~`c_wetday_p99_mm`~~

~~`c_rx5_rl2_mm`~~, ~~`c_rx5_rl25_mm`~~ (all c_rx5_rl10_mm)

X: land cover and terrain

~~`urban_share`~~, ~~`developed_share`~~ (impervious_mean)

~~`water_share`~~, ~~`water_km2`~~ (hand_lt5_share)

~~`elev_mean`~~, ~~`slope_mean`~~, ~~`flat_share`~~, ~~`hand_mean`~~ (hand_lt5_share)

A: wetlands

~~`wetland_change_since_1999`~~ (wetland_share)

In the panel, not used yet

Census and ACS: X

`housing_units`, `population`, `median_value`

Census and ACS: A candidates

`median_income`, `poverty_rate`

`mobile_home_share`, `built_pre1980_share`

Planned, not built

`policies_in_force`, `buyouts_cum`, `elevations_cum` (A)

`soil_cd_share` (X)

The outcome columns (`PropertyDmgADJ`, `log_PropertyDmgADJ`, `has_damage`) and keys (`GEOID`, `year`, `split`) are not inputs. Labels follow HANDOFF.md: C is rain climate; terrain is susceptibility and sits in X.